



2501- ELL 1000
“Introduction to Electrical Engineering”

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GROUP NO.	6	5
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Lab Report for Experiment 2

*Verification of Network Theorems:
Kirchhoff's Current Law (KCL), Kirchhoff's Voltage Law (KVL),
Thevenin's and Norton's Theorems*

31/08/2025

• Readings of Resistance Values measured from Colour Bands and Multimeter:

This experiment verifies fundamental network theorems in DC circuits — Kirchhoff's Current Law (KCL), Kirchhoff's Voltage Law (KVL), and the Thévenin and Norton equivalence theorems. Using standard resistors identified by colour codes and confirmed with a digital multimeter, we assembled a simple network, predicted node voltages and branch currents from theory learnt in class, and then compared them with measured values. KCL is validated by showing the algebraic sum of currents at a node is zero, while KVL is confirmed by showing that the sum of potential differences around closed loops equals the source voltage in the circuit.

Resistor	Value of Resistances read from colour codes with the units	Tolerance (if that band is visible / present)	Value of Resistance measured by the Multimeter	Do the values match (use a multimeter)?
R1	4k6	120(5)	4k72	Yes
R2	4k6	-20(5)	4k58	Yes
R3	1k0	-20(5)	0k98	Yes
R4	10k0	+80(5)	10k08	Yes
R5	1k0	-10(5)	0k99	Yes
R6	1k0	-20(5)	0k98	Yes
R7	4k6	+100(5)	4k7	Yes
RL	1k0	-20(5)	0k98	Yes

Table 1: Resistance measurement Table

NOTATIONS:

- $akbc = a.bc \text{ k}\Omega$
- In Tolerance: $\pm a(b) = a \text{ }\Omega(\pm b \text{ percent})$

Table: Reading the Resistance values from Colour Bands

	Value of Resistances read from colour codes with the units	Tolerance	Value of Resistance measured by the Multimeter	Do the values match
R1	$46 \times 10^2 \Omega$	$+120\Omega (\pm 5\%)$	$4.72 \times 10^3 \Omega$	Yes
R2	$46 \times 10^2 \Omega$	$-20\Omega (\pm 5\%)$	$4.58 \times 10^3 \Omega$	Yes
R3	$10 \times 10^2 \Omega$	$-20\Omega (\pm 5\%)$	$0.98 \times 10^3 \Omega$	Yes
R4	$10 \times 10^3 \Omega$	$+80\Omega (\pm 5\%)$	$10.08 \times 10^3 \Omega$	Yes
R5	$10 \times 10^2 \Omega$	$-10\Omega (\pm 5\%)$	$0.99 \times 10^3 \Omega$	Yes
R6	$10 \times 10^2 \Omega$	$-20\Omega (\pm 5\%)$	$0.98 \times 10^3 \Omega$	Yes
R7	$46 \times 10^2 \Omega$	$+100\Omega (\pm 5\%)$	$4.70 \times 10^3 \Omega$	Yes
RL	$10 \times 10^2 \Omega$	$-20\Omega (\pm 5\%)$	$0.98 \times 10^3 \Omega$	Yes

ADH
19/09/25

● Measurement of Voltage Drop and Current across each Resistor:

This table records the measured voltage drop (V_k) and branch current (I_k) for each resistor in the network and the load at three DC supply levels: 5 V, 10 V, and 15 V. For each supply setting, the first row captures the voltage across each component, and the second row captures the corresponding current through that component, measured with a multimeter. These data are used to check KCL at the nodes, and validate KVL around loops.

DC Voltage Source V_{DC}	V and I measurement across Resistance R_k ; $k = 1..7$	For R_1	For R_2	For R_3	For R_4	For R_5	For R_6	For R_7	For R_L
5V	V(volts)	3.57	0.22	0.74	0.47	0.09	0.07	0.08	0.64
5V	I(mA)	1	0	0	0	0	0	0	0
10V	V(volts)	7.21	0.97	1.49	0.34	0.14	0.14	0.14	1.28
10V	I(mA)	1.57	0	1	0	0	0	0	1
15V	V(volts)	10.82	1.45	2.24	0.48	0.26	0.21	0.21	1.93
15V	I(mA)	2.35	0.31	2.2	0	0.2	0.2	0	1.9

Table 2: V and I measurement Table

Table 2: Verification of KVL

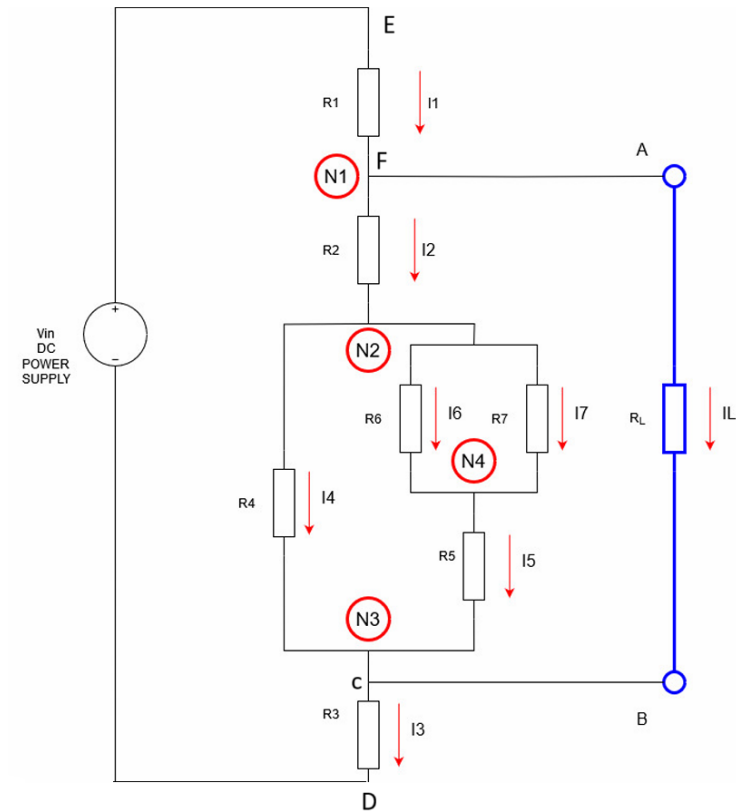
DC Voltage (V_{DC})	V & I measured across resistance R_k ; ($k=1,2,...,7$)	For R_1	For R_2	For R_3	For R_4	For R_5	For R_6	For R_7	For R_{load}
5V	V(volts)	3.57	0.22	0.74	0.47	0.09	0.07	0.08	0.64
5V	I(mA)	1	0**	0**	0**	0**	0**	0**	0**
10V	V(volts)	7.21	0.97	1.49	0.34	0.14	0.14	0.14	1.28
10V	I(mA)	1.57	0**	1	0**	0**	0**	0**	1
15V	V(volts)	10.82	1.45	2.24	0.48	0.26	0.21	0.21	1.93
15V	I(mA)	2.35	0.31	2.2	0*	0.2	0.2	0*	1.9

0** signifies very low I due to very high R value.
(not measurable)

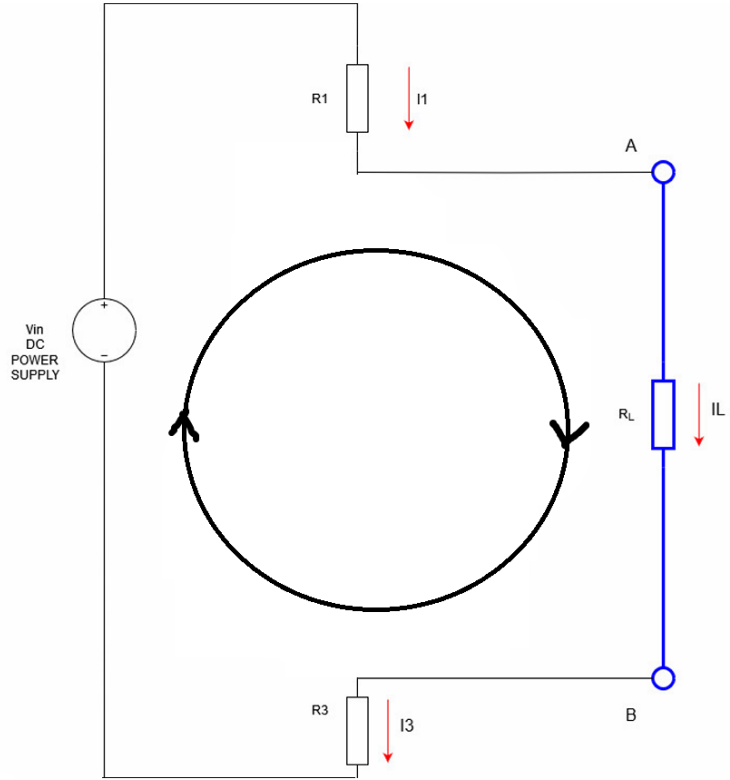
4 bits 15/02/25

- Verification of Network Theorms:

KVL :- For loop ABCDEFA: $\Sigma V_k = 0.01V$ for $V_{app} = 15V$



Provided Circuit



Circuit showing loop on which KVL is being applied

KCL :-

Node	ΣI_K (mA)
N_1	0.14
N_2	0.11
N_3	2
N_4	0

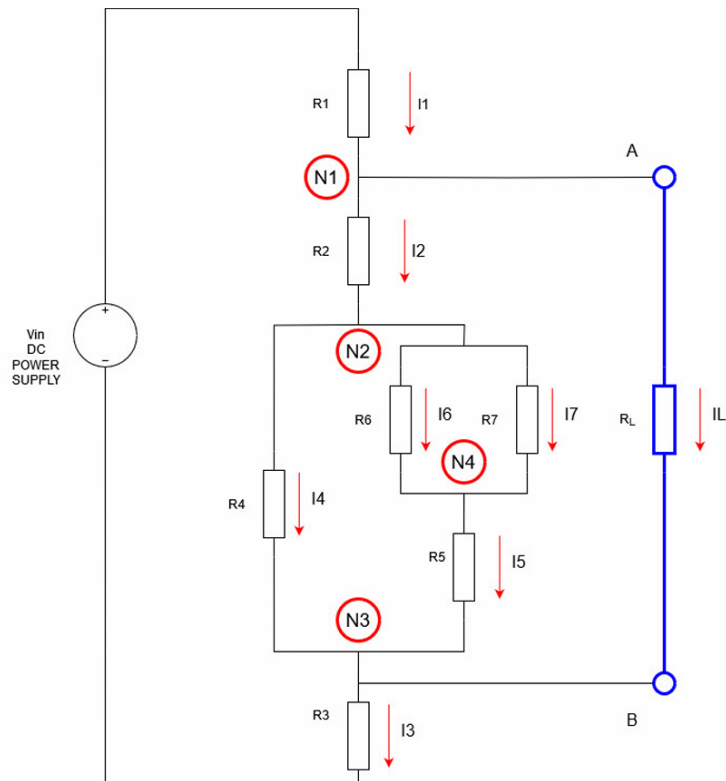
Table 3: Table showing application of KCL across every node

The errors we have obtained may be due to the following reasons:

1. Instrument inaccuracy
2. Lead/contact resistance and poor connections
3. Resistor tolerance and component variation
4. Offset/zero error of instruments
5. Noise, interference

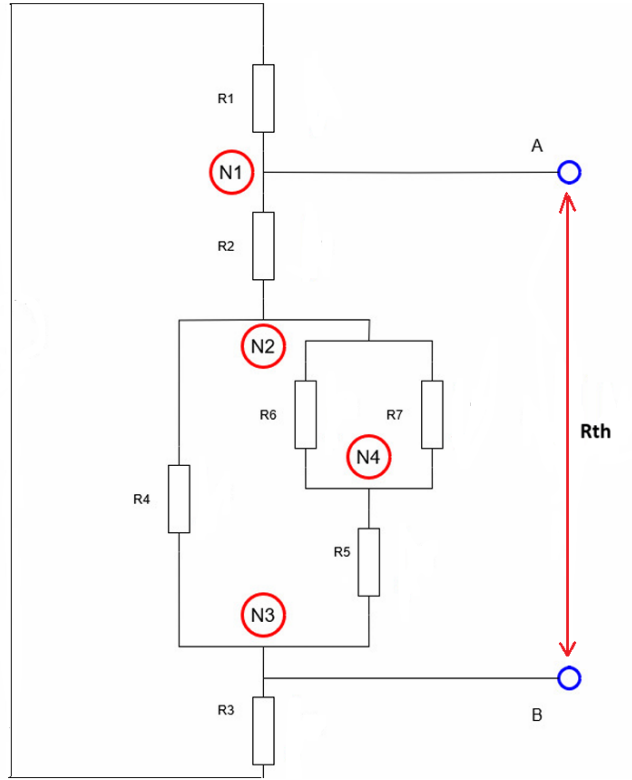
• Verification of Thevenin's Theorem:

We began with the given circuit consisting of a voltage source and a network of resistors. Our first step was to identify the output terminals where the load resistor would be connected.



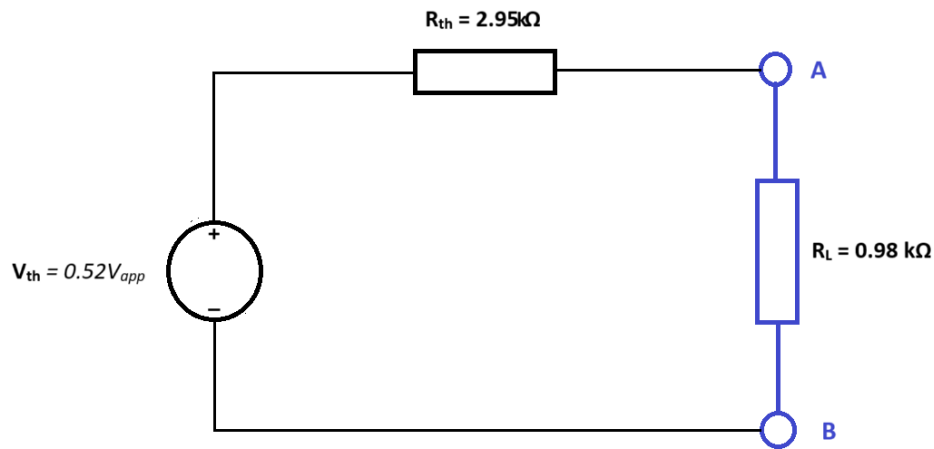
Provided Circuit

To find R_{th} , we removed the load resistor and “turned off” the voltage source (shorted it, since it was ideal). Looking back into the circuit, we reduced the series-parallel resistors to a single value, which gave us R_{th} .



Calculation of R_{th}

Finally, we drew the Thevenin equivalent circuit: a single V_{thev} in series with R_{thev} . We then connected two different load resistors one by one and measured the voltage and current in each case. The results matched the original circuit closely, confirming the correctness of our Thevenin model.

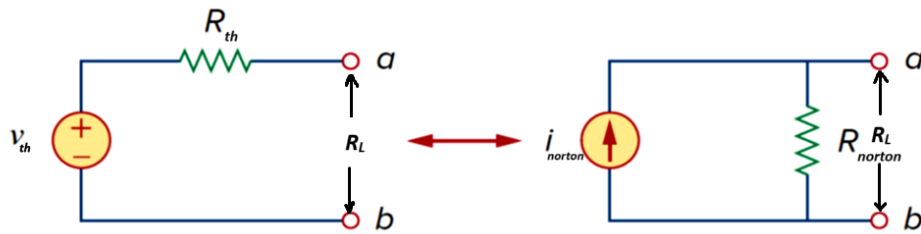


V_{app}	V_{th}
5V	2.6V
10V	5.2V
15V	7.8V

Thevenin's equivalent Circuit

● Verification of Norton's Theorm:

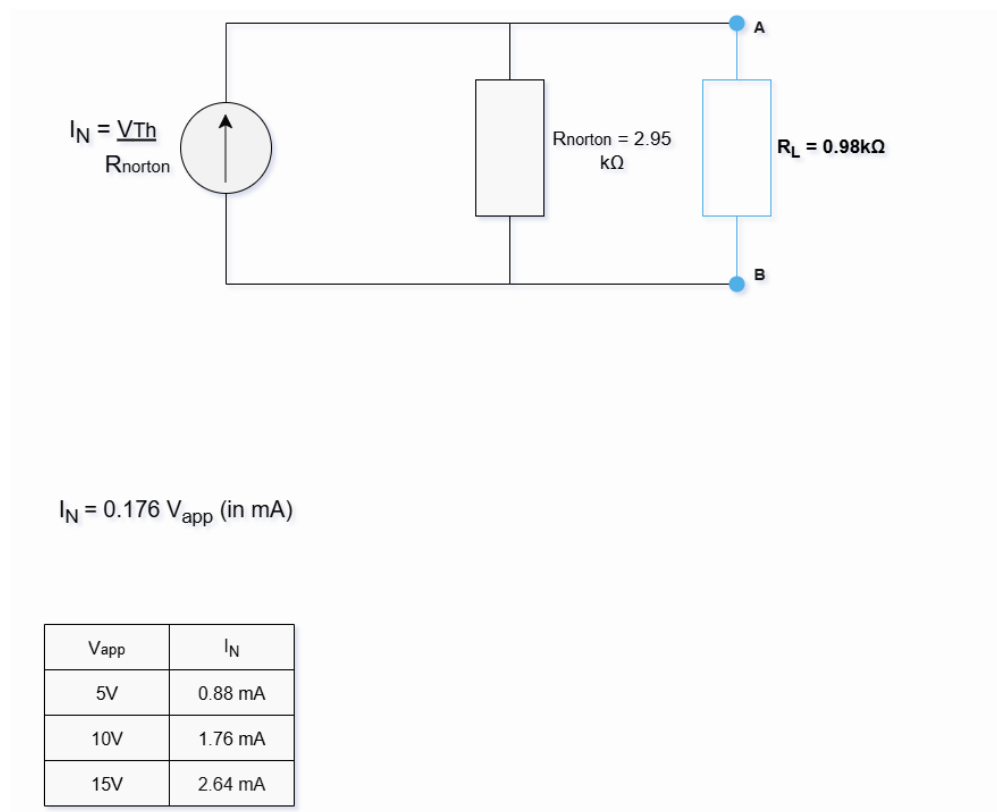
We started by finding the Norton current (I_{Norton}). For this, we converted the thevenin's equivalent cicuit's voltage source into a equivalent current source by below means.



Thevenin's to Norton's Conversion

We know that $R_{th} = R_{norton}$ as Both Thevenin's Theorem and Norton's Theorem are used to generate equivalent simpler circuits when we analyse circuits. Both generate the same equivalent resistance when 'looking into' the two terminals of the circuit

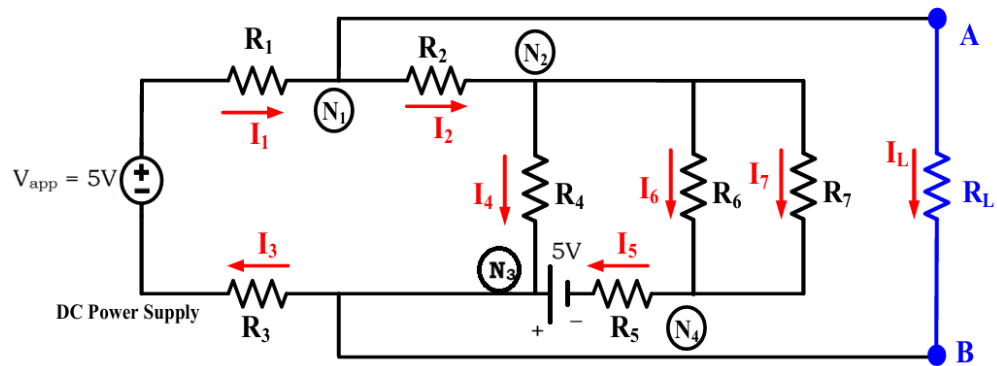
So, we get the Norton's equivalent circuit as:



Norton's equivalent Circuit

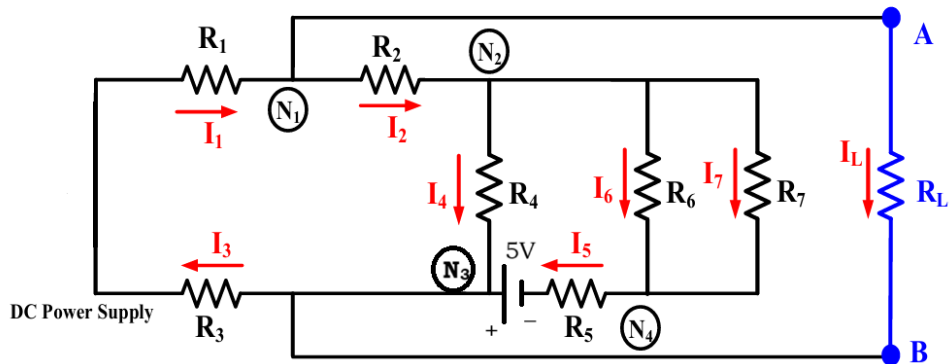
● Verification of Superposition principle:

We started with the given circuit containing two independent sources. In our case, the first source was V_1 and the second was V_2 (though a current source could also be used).

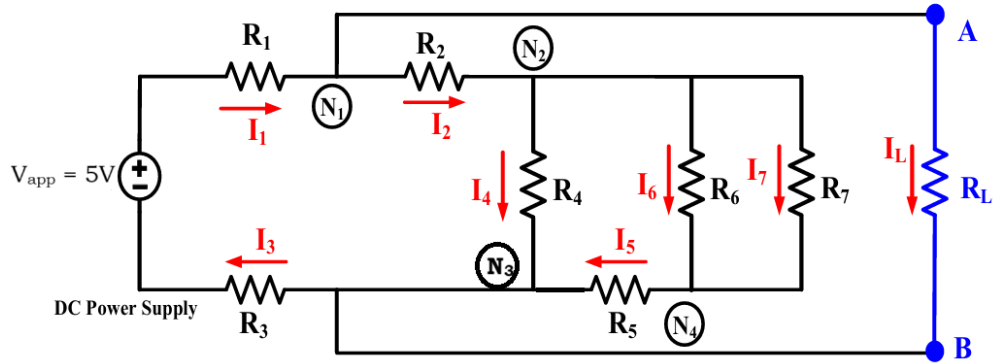


Circuit with both sources V_1 and V_2
 $V_1 = V_{app}$ and $V_2 = \text{second voltage source}$

Now, we removed the second voltage source



Now, we removed the applied voltage source



Voltage (and current) across R2 due to Source 1 alone = $V_k(1) = 1.67V$, $I_k(1)=0.36mA$
 Voltage (and current) across R2 due to Source 2 alone = $V_k(2) = 1.07V$, $I_k(2) = 0.23mA$
 Superposing, Voltage (and current) across R2 due to Source 1 and Source 2 together = $V_k(1) + V_k(2) = 2.74V$, $I_k(1) + I_k(2) = 0.59mA$ while the measure value of $V_k(1) + V_k(2) = 2.74V$, $I_k(1)+I_k(2)=0.6mA$